

23 Pricing and financing

13 primary questions | Oldest to newest

2026 objectives: 4.3

19 related questions are filed elsewhere in this volume. Find them by selecting this chapter and enabling related topics in [index.html](#).

Click a question below to begin.

CA1-P2 April 2005 Q7	23 marks
CA1-P2 September 2005 Q3	18 marks
CA1-P2 September 2005 Q5	34 marks
CA1-P2 September 2008 Q5	16 marks
CA1-P2 April 2009 Q4	11 marks
CA1-P2 April 2009 Q5	19 marks
CA1-P2 April 2010 Q7	28 marks
CA1-P2 September 2010 Q1	8 marks
CA1-P2 April 2011 Q6	21 marks
CA1-P2 September 2014 Q4	13 marks
CA1-P2 April 2015 Q7	28 marks
CA1-P2 April 2017 Q6	17 marks
CP1-P2 April 2021 Q2	58 marks

CA1-P2 | April 2005 | Q7

Current topic | 23 marks total | Original PDF p4

Related: Ch 35, Ch 1.

7 A small company is negotiating a remuneration package with an individual aged 50 who it wishes to appoint as its new chief executive officer. The individual has asked the company to guarantee him a pension of \$200,000 a year from age 60. The company is considering how it might finance this benefit.

- (i) List four approaches to the timing of the company's payments to meet this benefit, and outline how any surpluses or deficits could be removed under each approach. [5]

The company decides to provide the benefit without contributing to a specific fund in advance.

- (ii) Discuss the effect of this on the amount of capital that the company requires in order to operate its business. [7]

The company then decides that it cannot accept the risk of the guaranteed pension. Instead it is agreed that the individual will receive a higher salary and will make his own pension arrangements. The individual asks for advice on the amount of contributions he should pay to meet the target pension of \$200,000 a year.

- (iii) Describe the additional information required from the individual in order to enable an advisor to give suitable advice. [7]
- (iv) Outline the points that would be made when presenting the advice (without carrying out any calculations). [4]
- [Total 23]

CA1-P2 | September 2005 | Q3

Current topic | 18 marks total | Original PDF p2

Related: Ch 18, Ch 26.

3 A general insurance company has been asked to quote premium rates by a large country-wide motor dealer for residual value insurance on loans arranged by the motor dealer for car purchase. The loans would typically have a term of 5 years. The insurance is offered at point of sale of a car to the car buyer. The cover provided is the difference between the amount required to pay off the loan and the amount recoverable under the policyholder's motor insurance policy if a car is written off as a total loss through accident or theft.

- (i) In order for the insurance company to determine the premium rates to be quoted:
- (a) List the information required.
 - (b) Describe an approach to determining and expressing the premium rates that could be used.
- [12]
- (ii) State the risks involved for the insurance company in providing this benefit and suggest how these could be mitigated.
- [6]
[Total 18]

CA1-P2 | September 2005 | Q5

Current topic | 34 marks total | Original PDF p3

Related: Ch 17, Ch 19.

- 5**
- (i) State the requirements and basic features of an actuarial model to be used to determine the premium rates for a new life insurance product. [7]
- (ii) Discuss the relative importance of the assumptions for:
- investment return
 - mortality experience
 - expenses
 - withdrawal rates
- when setting the pricing basis for the following products:
- (a) a without profits term assurance
- (b) a without profits level immediate annuity [12]

For each product in part (ii), it has been decided that the pricing model should involve stochastic simulation of some of the four items listed above.

- (iii) For each product, state with reasons which of the four items would be modelled stochastically. [6]
- (iv) Give reasons why the prices charged for the products might be different from the prices determined by the modelling process, and discuss the implications for the company of such differences. [9]
- [Total 34]

CA1-P2 | September 2008 | Q5

Current topic | 16 marks total | Original PDF p2

Related: Ch 18.

- 5 (i) (a) Explain the term “underwriting cycle”.
- (b) Define the term “risk premium”, and state the general formula that may be used to calculate a risk premium. [4]

A large general insurance company writes warranty business for construction companies that build housing. The warranties are for a period of twelve years and cover major damage caused by building defects in new homes. The company has been writing this business for the last ten years.

CA1-P2 | September 2008 | Q5 - continued

Current topic | 16 marks total | Original PDF p3

Related: Ch 18.

- (ii) Outline the main data issues that will need to be considered when calculating a theoretical risk premium for this business. [2]
- (iii) Outline the factors (other than the theoretical risk premium) that should be taken into account when determining the actual premium to be charged for this business. [6]

A particular construction company has requested that its premium rates be reviewed.

- (iv) Discuss the further considerations that the insurance company will take into account when responding to this request. [4]
- [Total 16]

CA1-P2 | April 2009 | Q4

[Current topic](#) | [11 marks total](#) | [Original PDF p2](#)

Examiner report and original paper: use the links for this question in [index.html](#).

- 4** An individual, whose motor insurance policy is due for renewal, uses a website to compare quotes from a range of providers including his current insurer.

The individual supplies personal and other details and specifies the type of cover required. Based on this information, the website gives a wide range of premiums, all of which are lower than the premium currently being paid.

Explain possible reasons for the range and size of the results of this comparison exercise. [11]

CA1-P2 | April 2009 | Q5

Current topic | 19 marks total | Original PDF p3

Related: Ch 22, Ch 29.

- 5** In twelve months' time, a country is due to hold an election to choose its government. There are likely to be five parties contesting the election. A respected polling organisation currently estimates the probability of each party winning the election to be:

<i>Party</i>	A	B	C	D	E
<i>Probability %</i>	40	30	15	10	5

A company wishes to use this data to construct an investment product linked to the result of the election.

In return for a single premium, investors will be able to select the party that they think will win. If their selection does win, they will receive a fixed multiple of their original investment. If their selection does not win, the investment is lost. The multiples offered vary between each party and for each party they will vary over time depending on the relative probabilities of victory at each point in time.

The multiples can be non-integral, for example 4.333 or 2.375 per unit invested

Twelve-month money market interest rates are currently 5%.

- (i) Calculate, stating the assumptions you use, the multiples that the company should currently offer in respect of each party winning the election. [6]
- (ii) Discuss the issues relating to the determination of the election result that will need to be clarified when drawing up the terms of the contracts. [3]

Six months later, two of the parties withdraw from the election.

- (iii) State the actions the company could take in respect of investments already made on each of the original five parties. [2]

A rival company offers the following terms on the three remaining parties:

<i>Party</i>	X	Y	Z
<i>Multiple</i>	4.0	2.5	3.5

- (iv) Explain why these terms could cause problems for this rival company. [2]

Many companies offer similar products using the results of a range of events as the underlying investment on which contracts are based. Primarily, these products involve predicting the outcomes of individual or an accumulation of sporting contests. Most results are known and winning payouts made on the day, or within a few days, of the investment being made.

- (v) Explain how the concept of reinsurance could be used to manage the risk inherent in these products. [6]

[Total 19]

CA1-P2 | April 2010 | Q7

Current topic | 28 marks total | Original PDF p4

Related: Ch 22, Ch 3.

- 7** A charity that exists to promote the interest of elderly people has prepared a report on the cost of travel insurance. The findings are that people aged over 65 are charged, on average, 80% more than people under age 65 for similar travel insurance products. The charity argues that this is a clear case of age discrimination and profiteering by insurance companies. It is pressing the government to introduce legislation that will require insurers to charge the same amount for the same policy to all policyholders irrespective of age.

To support its case, the charity points out that the vast majority of claims under travel insurance policies are for events such as cancellations, delays, loss of luggage, theft of property or inadequate standards of service or accommodation. It argues that these issues are not dependent on the age of the policyholder and so elderly people should not be charged more.

The charity states that elderly people do not engage in high risk behaviour or dangerous activities such as skiing, whereas younger people's travel tends to involve more risk and hence they are more likely to claim.

A representative of the insurance industry has been asked by a prominent media organisation to make a statement and answer questions in response to the report.

- (i) Outline the introductory remarks that the representative could make to explain the influences on the travel insurance market and the insurance industry's approach to elderly travellers. [5]
- (ii) Discuss the specific points the representative should make when analysing the charity's report and justifying current industry practice. [9]

The marketing director of a large insurance company believes that the charity's report presents an opportunity. The director argues that by introducing a travel insurance policy where premiums are the same for all ages, the company's market share can be increased significantly.

- (iii) Outline the risks to the insurance company inherent in following such a strategy. [7]
- (iv) Describe how the insurance company could structure and market a travel insurance product that does not involve higher premiums for elderly people but could control the risks in (iii). [7]

[Total 28]

CA1-P2 | September 2010 | Q1

[Current topic](#) | [8 marks total](#) | [Original PDF p2](#)

Examiner report and original paper: use the links for this question in [index.html](#).

- 1** An insurance company is considering introducing a health insurance policy, which will cover the cost of medical treatment. Premiums will be payable annually.
- The policy will incorporate a guarantee such that premiums will be maintained at the existing rate if no claims have been made in the previous year.
- Discuss the main considerations that will need to be taken into account when determining the appropriate premium to charge. You do not need to consider issues relating to expenses or investment returns and strategy. [8]

CA1-P2 | April 2011 | Q6

Current topic | 21 marks total | Original PDF p3

Related: Ch 11, Ch 38.

- 6** Over the last three months, the government of a large developed country has cut the rate of interest charged on the very short-term loans that it makes to the banking sector from 6% pa to 1% pa. Yields on other money market instruments have fallen in a similar way.

- (i) Discuss how this change may affect the interest rates charged on:
- (a) Mortgage loans used to purchase domestic residential property.
 - (b) Balances outstanding on individual customers' credit card accounts.
- [10]

A large financial institution has a significant credit card business covering a wide range of personal customers in its domestic market. Different customers are charged different interest rates appropriate to their circumstances. Following the changes in short-term interest rates, the institution is reviewing the interest rates it charges to its credit card customers.

- (ii) Describe the process the credit card company would use to establish a revised interest rate charging structure. [6]
- (iii) Discuss how the company will assess the effect of variations in experience. [5]
- [Total 21]

CA1-P2 | September 2014 | Q4

[Current topic](#) | [13 marks total](#) | [Original PDF p2](#)

Related: Ch 18, Ch 19.

- 4** A general insurance company intends to launch a new product. It will cover policyholders for medical costs arising from an accident.

Describe how to determine the premiums for this product, including the process of collecting data and setting pricing assumptions. [13]

CA1-P2 | April 2015 | Q7

Current topic | 28 marks total | Original PDF p4

Related: Ch 18, Ch 30.

- 7** An insurance company is to provide insurance for passengers travelling through a new city airport which opened six months ago.

The premium for the insurance policy will be paid by the airport at no cost to passengers. This is an incentive to encourage passengers to travel through this airport, rather than travelling through nearby alternative airports which have been established for some time.

The airport has considered the risks which could be covered by the insurance policy and decided that the following risks will be covered:

1. Indemnity for the loss of, or damage to, luggage for every passenger, if it occurs at the airport, up to a maximum payout of \$10,000 for each passenger.
 2. A payout of up to \$100,000 on death or injury if there is an aircraft “incident” on flights arriving at, or departing from, the airport.
- (i) List other risks which could have been covered under this insurance policy. [3]
 - (ii) Describe possible sources of data, and their limitations, for the risks of luggage being lost or damaged. [6]
 - (iii) Outline how the insurance of luggage being lost or damaged could be priced, assuming that detailed credible data was available. [10]
 - (iv) Discuss reasons why providing this insurance cover may not encourage passengers to fly from the airport. [4]
 - (v) Explain how the low likelihood risk of a plane with over 300 passengers on board crashing at the airport could be managed by the insurance company. [5]
[Total 28]

CA1-P2 | April 2017 | Q6

Current topic | 17 marks total | Original PDF p3

Related: Ch 35, Ch 14.

- 6** A sports club from a small town has, for the first time in its history, reached the later stages of a high profile national cup competition.

The remainder of the competition consists of seven well attended rounds and a prestigious final at the country's major sports stadium.

- (i) Set out the various sources of the revenues the club may receive from its continued participation in the competition. [2]

The club have decided to securitise part of their future revenues. Three quarters of the revenues from specified sources will be given to investors as and when they arise. In return, the investors will pay the club a cash sum before the next stages of the competition start.

- (ii) Explain why the club may have chosen this approach and the risks it may face. [6]

- (iii) Discuss how the price to be charged to investors would be determined. [7]

- (iv) Identify the type of investor that this opportunity might appeal to. [2]
[Total 17]

CP1-P2 | April 2021 | Q2

Current topic | 58 marks total | Original PDF p4

Integrated case study. The full scenario is retained under one primary chapter; related chapters identify its other substantial topics. Related: Ch 18, Ch 26, Ch 30, Ch 12, Ch 14, Ch 38.

2 Background

The government maintains and operates the railway system for the whole of a developed country. The government is responsible for ensuring that trains run on time and ticket prices provide customers with value for money. The train system is critical for most customers who commute to work in the main cities in the country.

The government has the following policies for refunding customers if a train journey is delayed or cancelled:

- 15–29 minutes late = 20% of the customer's ticket price refunded
- 30–59 minutes late = 40% of the customer's ticket price refunded
- 60+ minutes late or train cancelled = 100% of the customer's ticket price refunded.

The government also maintains the roads in the country and has introduced a new traffic charge for the country's six largest cities. The traffic charge is applied to vehicles entering or moving within each city traffic charging zone between 8 am and 10 am and between 4 pm and 7 pm on weekdays (excluding national holidays). The traffic charge is inflation-linked increasing each year. The government expects to keep the traffic charge and inflation-link policy in place for at least the next 10 years.

The introduction of the new traffic charge has not been popular with citizens. Due to the poor punctuality of the train service, the introduction of the traffic charge has had a minimal impact on the number of vehicles entering the six largest cities.

The government is planning to outsource the maintenance of the railways to an overseas company, with the expectation that this will significantly improve the reliability and punctuality of trains. In addition, the government is evaluating the following package of proposals:

- | | |
|----------|--|
| Action 1 | To insure the risk of the train services being delayed/cancelled: in return for a premium, an insurance company would pay the necessary refunds on tickets. |
| Action 2 | To sell the future rights to the traffic charge income to a pension scheme: in return for a one-off payment, the pension scheme would receive all future income from the traffic charge. The government is willing to split the rights into individual cities if more than one pension scheme is willing to invest. The rights would be sold using a bidding system to maximise the price. |

Questions

- (i) Outline the risks of Action 1 from an insurance company's viewpoint. [5]
- (ii) Discuss the data an insurance company would need in order to price the contract under Action 1, including commenting on the limitations of the data. [10]
- (iii) Discuss the factors an insurance company would need to consider (other than the availability of data) when pricing the policy in Action 1. [12]

CP1-P2 | April 2021 | Q2 - continued

Current topic | 58 marks total | Original PDF p5

Integrated case study. The full scenario is retained under one primary chapter; related chapters identify its other substantial topics. Related: Ch 18, Ch 26, Ch 30, Ch 12, Ch 14, Ch 38.

- (iv) Describe the risk management tools available to an insurance company should it offer a contract under Action 1. [7]
 - (v) Outline the reasons a pension scheme may want to invest in the roads where the new traffic charge will apply. [5]
 - (vi) Outline the advantages and disadvantages of using a discounted cashflow method to value the future traffic charge rights. [2]
 - (vii) Discuss the issues that the pension scheme will need to consider in determining the cashflows to value the future traffic charge rights under Action 2. [6]
 - (viii) Discuss the risks the pension scheme should consider before submitting a bid under Action 2. [8]
 - (ix) Suggest how the government could evaluate whether the package of proposals is a success. [3]
- [Total 58]